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Causal Counterfactual Policy Ranking: Beyond Expected Value in Off-Policy Evaluation

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Abstract

Off-policy evaluation (OPE) is the standard approach for selecting among candidate policies in reinforcement learning without further environment interaction. Existing methods focus on estimating absolute policy values, yet practical deployment requires *reliable ranking*: a method can have low value-estimation error and still select the wrong policy. We identify this *policy misranking* problem, characterize it theoretically, and propose a principled fix.

Our approach, **Probability of Regret (PoR)**, compares policies at the *matched-pair trajectory level*: $\mathrm{PoR}(\pi_A, \pi_B) = \Pr(G_A(\tau) > G_B(\tau))$. We extend PoR to sequential MDPs via a model-based estimator using Fitted Q-Evaluation (FQE) that completely avoids importance-sampling weights. We prove four theorems: (i) an IS misranking bound $\leq 2e^{-n_{\mathrm{eff}}\Delta^2/2C^2}$ that reveals the failure regime; (ii) a MARL impossibility theorem showing that opponent distribution shift cannot be corrected offline; (iii) a PoR misranking bound $\leq e^{-2n_{\mathrm{eff}}\rho^2}$ that is independent of the return range C; and (iv) a dominance-selection stability guarantee.

We validate across a **five-environment benchmark** (gridworlds of sizes 3×3 through 10×10, including stochastic transitions) under 25 distinct settings. PoR-Model achieves Spearman $\rho = 1.000$ in all 25 settings where IS/WIS rankings collapse to $\rho \approx -0.2$. Ablation experiments reveal a **100× sample-efficiency advantage**: PoR-Model correctly ranks policies with $N = 50$ trajectories, whereas WIS requires $N \approx 5000$. Code is available at github.com/dhruvgoyal/causal-rl.

1 Introduction

Reinforcement learning (RL) policies are increasingly deployed in high-stakes domains — clinical decision support, autonomous navigation, content recommendation — where experimenting with a new policy online is either expensive or ethically impermissible [17, 22]. Off-policy evaluation (OPE) addresses this by estimating a policy's performance from historical data collected under a different *behavior policy* [25, 28].

The misranking problem. In practice, OPE is used to *select* the best of K candidate policies. This reduces to a *ranking* problem: given estimates $\{\hat{V}(\pi_k)\}_{k=1}^{K}$, select $\arg\max_k \hat{V}(\pi_k)$. A critical but underappreciated failure mode is that a method may estimate values inaccurately *yet preserve the correct ranking*, or conversely produce low estimation error while reversing the true ordering. Hand [8] coined the term "classifier illusion" for the analogous phenomenon in machine learning: optimizing a single aggregate metric conceals that the "best" model systematically underperforms on the subpopulation that matters. We identify the OPE analog: **policy misranking**, and show it is far more common than previously recognized.

Why standard OPE misranks. Standard IS and WIS estimators compare policies by their *separately estimated average values*. Three structural pathologies cause misranking:

1. **ESS collapse.** Trajectory IS weights $w^{\pi}(\tau) = \prod_t \pi(a_t|s_t)/\beta(a_t|s_t)$ become exponentially large or small as the horizon grows and the behavior policy diverges from the target. The effective sample size $\mathrm{ESS} = (\sum_i w_i)^2/\sum_i w_i^2$ collapses from N to near 1, turning the estimate

into a single-trajectory dominated quantity. IS estimates then range over $[10^{-52}, 10^{28}]$ in our experiments — numerically meaningless yet sometimes accidentally rank-preserving.

2. **Zero-weight problem.** For near-deterministic target policies, $w^\pi(\tau) \approx 0$ for *almost every* trajectory under an exploratory behavior policy. Matched-pair IS comparison $G_i w_A^i > G_i w_B^i$ then collapses to comparing near-zero against near-zero, yielding arbitrary rankings.
3. **MARL opponent shift.** In multi-agent environments, the value of policy π^i depends on who the opponents are. Data collected against opponent pool A carries no information about pool B, making cross-pool ranking provably impossible without access to the new opponents' policy (Theorem 3).

Our solution: causal counterfactual dominance. Instead of ranking policies by their separately estimated expected values, we rank by **Probability of Regret (PoR)** [14]:

$$\mathrm{PoR}(\pi_A, \pi_B) = \Pr_{\tau \sim \beta}[G_A(\tau) > G_B(\tau)],$$

where $G_A(\tau)$, $G_B(\tau)$ are the counterfactual returns that π_A, π_B *would have obtained in the same context* τ. This matched-pair causal comparison eliminates common trajectory noise. For sequential decisions, we estimate PoR via Fitted Q-Evaluation (FQE) [4], bypassing IS weights entirely (Section 7).

Contributions. We make the following contributions:

1. **Theory (Theorems 1–5)**: Four formal results establishing misranking bounds, a MARL impossibility result, a tighter bound for PoR that is free of the return range C, and a dominance-selection stability guarantee. Full proofs with lemmas are in Appendix A.
2. **PoR-Model (Section 7.2)**: A model-based PoR estimator via FQE that circumvents all three pathologies above. Algorithm 1 is self-contained and straightforward to implement.
3. **Five-environment benchmark (Section 9)**: Evaluation spanning gridworlds with 9–100 states, dense and stochastic rewards, and $\varepsilon_\beta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. PoR-Model achieves $\rho = 1.000$ in all 25 settings; IS/WIS achieve $\rho \leq -0.2$ at worst coverage.
4. **100× sample efficiency (Section 11)**: Ablation confirms PoR-Model correctly ranks with $N = 50$ trajectories; WIS requires $N \approx 5000$ under the same bad coverage.

2 Background

MDP and offline dataset. Let $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ be a Markov Decision Process with finite state space $|\mathcal{S}| = S$, action space $|\mathcal{A}| = A$, transition kernel P, reward R, and discount $\gamma \in [0, 1)$. We observe a fixed dataset $\mathcal{D} = \{\tau_i\}_{i=1}^N$, $\tau_i \sim \beta$, collected under behavior policy β. Trajectory $\tau_i = (s_0^i, a_0^i, r_0^i, \ldots, s_T^i)$; the discounted return is $G(\tau_i) = \sum_{t=0}^T \gamma^t r_t^i$.

IS family of estimators. The trajectory importance ratio is $w^\pi(\tau) = \prod_{t=0}^T \pi(a_t|s_t)/\beta(a_t|s_t)$.

$$\hat{V}_{\mathrm{IS}}(\pi) = \tfrac{1}{N}\textstyle\sum_i G_i w_i^\pi \quad \text{(unbiased, high variance)} \tag{1}$$

$$\hat{V}_{\mathrm{WIS}}(\pi) = \textstyle\sum_i G_i w_i^\pi \big/ \sum_j w_j^\pi \quad \text{(biased } O(1/N)\text{, self-normalized)} \tag{2}$$

$$\hat{V}_{\mathrm{PDIS}}(\pi) = \tfrac{1}{N}\textstyle\sum_i \sum_t \gamma^t r_t^i \prod_{s \leq t} \pi(a_s^i|s_s^i)/\beta(a_s^i|s_s^i) \quad \text{(per-decision IS)} \tag{3}$$

Doubly robust (DR) methods [3, 11] add a model-based control variate to reduce variance. Marginalized IS [35] and DualDICE [21] correct stationary distribution shift rather than trajectory-level weights.

Effective sample size (ESS). $\mathrm{ESS}(\mathbf{w}) = \left(\sum_i w_i\right)^2 \big/ \sum_i w_i^2$, with $\mathrm{ESS}_{\mathrm{frac}} = \mathrm{ESS}/N \in [0, 1]$. When $\beta = \pi$, $\mathrm{ESS}_{\mathrm{frac}} = 1$; under severe shift, $\mathrm{ESS}_{\mathrm{frac}} \to 0$.

Fitted Q-Evaluation (FQE). FQE [4] estimates $Q(s, a)$ from offline data by iterating Bellman target updates:

$$Q^{(\ell+1)}(s, a) \leftarrow \mathbb{E}_{(s,a,r,s')\in\mathcal{D}}\left[r + \gamma \max_{a'} Q^{(\ell)}(s', a')\right]. \tag{4}$$

At convergence, the policy value at any initial state is $\hat{V}_\pi(s_0) = \sum_a \pi(a|s_0)Q(s_0, a)$.

3 Related Work

Off-policy evaluation. IS-based OPE dates to Precup et al. [25] and the classical Horvitz–Thompson estimator. Jiang and Li [11] and Dudík et al. [3] introduce doubly robust estimators that combine a model baseline with IS correction. Thomas and Brunskill [31] provide high-confidence OPE bounds; Thomas et al. [32] convert these to safe policy improvement guarantees. Liu et al. [18] handle infinite-horizon settings via a stationary distribution correction. Xie et al. [35] propose marginalized IS (MIS) that operates on state-action distributions rather than trajectories. Nachum et al. [21] reformulate OPE as minimax density-ratio estimation. Kallus and Uehara [13] develop double RL for semiparametric efficiency. Farajtabar et al. [5] improve DR robustness via shrinkage. The most comprehensive empirical comparison is Voloshin et al. [34], who find that no single method dominates across all distribution-shift regimes—a key motivation for our work. Uehara et al. [33] provide a recent theoretical survey.

Offline RL and benchmark datasets. Levine et al. [17] survey the offline RL landscape. Key benchmark datasets include D4RL [6] (locomotion, Antmaze, Adroit) and RL Unplugged [7] (Atari, control). Conservative methods [15] show that pessimism about out-of-distribution actions is essential for offline RL to work. Agarwal et al. [1] and Brandfonbrener et al. [2] explore when offline RL can succeed without explicit OPE. Model selection for offline RL is studied by Paine et al. [23] and Tang and Wiens [30], who show that cross-validation and OPE rankings are frequently inconsistent — echoing our misranking finding.

Counterfactual and causal OPE. Kawakami and Tian [14] introduce PoR in the bandit setting and prove that PoR induces a robust partial order. We extend their framework to sequential MDPs, prove tight bounds for the model-based estimator, and connect PoR to MARL impossibility. Muandet et al. [20] develop counterfactual mean embeddings for kernel-based causal inference. Swaminathan and Joachims [29] study counterfactual risk minimization in contextual bandits. Pearl [24] provides the foundational potential-outcomes framework underlying PoR's causal interpretation. Saito and Joachims [27] extend OPE to large action spaces via embedding, a complementary direction to our trajectory-level approach.

Multi-agent evaluation. Lowe et al. [19] (MADDPG) and Rashid et al. [26] (QMIX) are foundational multi-agent algorithms but assume full joint observability. Hernandez-Leal et al. [9] survey multi-agent RL broadly. To our knowledge, no prior work formalizes the impossibility of offline cross-pool evaluation in multi-agent settings (our Theorem 3).

4 Environments and Benchmarks

We evaluate on five tabular environments with known ground-truth values, enabling rigorous validation of ranking accuracy. All environments are parameterized by the same codebase; ground truth is obtained via on-policy Monte Carlo rollouts ($N_{\text{gt}} = 500$–2000 per policy).

Gridworld family (E1–E5). Each gridworld $\mathcal{G}_{m\times n}$ has mn states arranged in an $m\times n$ grid. The agent starts at $(0, 0)$ and must reach the goal at $(m-1, n-1)$. Actions: {up, right, down, left}. With slip probability p_{slip}, the executed action is replaced by a uniformly random action (E3 only). Rewards: $r = +1$ at goal, $r = -0.01$ per step.

Evaluation policies are ε-greedy with respect to a heuristic Q-function pointing toward the goal: $\varepsilon \in \{0.0, 0.1, 0.5, 0.8\}$. These span a wide performance range (from $V \approx 0.94$ to $V \approx 0.77$ in E1; from $V \approx 0.69$ to $V \approx -0.29$ in E5), making correct ranking non-trivial.

Table 1: Benchmark environment properties. S: state space size; Stoch.: stochastic transitions (slip probability); $\mathrm{ESS}_{\min}$: empirical minimum ESS fraction observed at $\varepsilon_\beta = 0.9$.

ID	Environment	S	A	Stoch.	T	$\mathrm{ESS}_{\min}$
E1	GW-3×3	9	4	0.00	50	∼10%
E2	GW-5×5	25	4	0.00	50	∼3%
E3	GW-5×5-S	25	4	0.15	50	∼3%
E4	GW-7×7	49	4	0.00	50	∼1%
E5	GW-10×10	100	4	0.00	50	<1%

ESS scaling. As state space size grows ($9 \to 100$), trajectory length must increase to reach the goal, causing the IS weight product to span more steps and ESS to collapse more severely. This creates a natural difficulty gradient: E1 is the easiest (high ESS even at $\varepsilon_\beta = 0.9$), E5 is the hardest (ESS$< 1\%$ at moderate mismatch).

Rock-Paper-Scissors (RPS) for MARL. Two opponent pools: Pool A (60% Rock, 30% Paper, 10% Scissors) and Pool B (10% Rock, 30% Paper, 60% Scissors), with total variation distance $d_{\mathrm{TV}}(\beta_A, \beta_B) = 0.5$. Three evaluation policies: AllRock, AllPaper, AllScissors. Ground truth win-rates computed by direct simulation (1000 matches).

5 Problem Formulation

Definition 1 (Policy ranking task)**.** Given $\mathcal{D} \sim \beta$ and K candidates $\Pi = \{\pi_1, \ldots, \pi_K\}$, recover the ordering σ^* such that

$$V(\pi_{\sigma^*(1)}) \geq V(\pi_{\sigma^*(2)}) \geq \cdots \geq V(\pi_{\sigma^*(K)}).$$

Definition 2 (Misranking and Spearman ρ)**.** π_A is *misranked above* π_B if $\hat{V}(\pi_A) > \hat{V}(\pi_B)$ yet $V(\pi_A) < V(\pi_B)$. The *Top-1 mismatch* is 1 if the estimated best policy differs from the true best. We measure overall ranking quality by Spearman's $\rho \in [-1, 1]$ between estimated and ground-truth orderings; $\rho = 1$ is perfect, $\rho = -1$ is a complete reversal.

Definition 3 (Probability of Regret [14])**.** For policies π_A, π_B and behavior β:

$$\mathrm{PoR}(\pi_A, \pi_B) \;=\; \Pr_{\tau \sim \beta}[G_A(\tau) > G_B(\tau)]\,, \tag{5}$$

where $G_k(\tau)$ is the counterfactual return that policy π_k would have obtained in context τ. Values above 0.5 indicate π_A stochastically dominates π_B at the trajectory level.

Definition 4 (Probability of Being Best)**.** $\mathrm{PoB}(\pi_k) = \Pr_{\tau\sim\beta}[k = \arg\max_j G_j(\tau)]$. The policy with the highest PoB is the one most likely to be best in a randomly drawn context.

The key insight is that $\mathrm{PoR} > 0.5$ is a *context-level* dominance statement: π_A beats π_B when they face the *same situation*. This is strictly stronger than $V(\pi_A) > V(\pi_B)$ under distribution shift, and addresses the misranking problem at its root.

6 Theoretical Analysis

We state all four results here; complete proofs with supporting lemmas are in Appendix A.

Assumption 1 (Overlap)*.* $\beta(a|s) > 0$ whenever $\pi_A(a|s) > 0$ or $\pi_B(a|s) > 0$, for all $s \in \mathcal{S}$.

Theorem 1 (IS Misranking Bound)**.** *Under Assumption 1, let $V_A > V_B$ with gap $\Delta = V_A - V_B > 0$, returns bounded in $[0, C]$, and $n^k_{\text{eff}} = N \cdot \mathrm{ESS}_{\text{frac}}(w^{\pi_k})$. Then:*

$$\Pr\Big(\hat{V}_{\mathrm{WIS}}(\pi_B) \geq \hat{V}_{\mathrm{WIS}}(\pi_A)\Big) \;\leq\; 2\exp\left(-\frac{\bar{n}_{\text{eff}}\,\Delta^2}{2C^2}\right), \tag{6}$$

where $\bar{n}_{\text{eff}} = \min(n^A_{\text{eff}}, n^B_{\text{eff}})$.

Corollary 2 (Sample complexity and failure regime)**.** *To achieve bound $\leq \delta$, require: $n_{\text{eff}} \geq 2C^2 \log(2/\delta)/\Delta^2$. The* failure regime *occurs when $n_{\text{eff}}\Delta^2/(2C^2) < 1$ (bound $> 2/e \approx 0.74$; ranking is essentially random). Under distribution shift $\mathrm{ESS}_{\text{frac}} \to 0$, so $N \to \infty$ is required; this is unavoidable for IS-based estimators (Appendix A.3).*

Theorem 3 (MARL OPE Impossibility Under Opponent Shift)**.** *In a two-agent game, let $\mathcal{D}$ be collected against opponent distribution β^{-i}. Any IS estimator for $V(\pi^i \mid \pi_B^{-i})$ (evaluation against new opponent π_B^{-i}) has irreducible bias:*

$$\text{Bias} \;\geq\; \left|V(\pi^i \mid \beta^{-i}) - V(\pi^i \mid \pi_B^{-i})\right| \;\geq\; 2\, d_{\text{TV}}(\beta^{-i}, \pi_B^{-i}) \cdot \Delta_{\min}, \tag{7}$$

where $\Delta_{\min}$ is a lower bound on the minimum advantage gap. When $d_{\text{TV}} = 1$, the bias can equal the full reward range.

Theorem 4 (PoR Misranking Bound)**.** *Under Assumption 1, let $\text{PoR}(\pi_A, \pi_B) > 0.5$ with dominance margin $\rho = \text{PoR}(\pi_A, \pi_B) - 0.5 > 0$. The PoR estimator satisfies:*

$$\Pr\Big(\widehat{\text{PoR}}(\pi_A, \pi_B) \leq 0.5\Big) \;\leq\; \exp\big(-2\, n_{\text{eff}}\, \rho^2\big)\,. \tag{8}$$

Remark 1 (Advantage of PoR over IS). (8) has *no dependence on C*: the comparison of $0/1$ indicators rather than scaled returns removes the return-range factor. By Corollary 9 (Appendix A), the PoR bound is tighter than the IS bound whenever $\rho > \Delta/(2C)$, which holds for well-separated policies. Under heavy-tailed IS weights (equivalently, large effective C), the IS bound collapses while PoR's remains tight.

Theorem 5 (Dominance Selection Stability)**.** *Let $W_k = \min_s V_k^s$ (worst-case value across scenarios). If all OPE errors are bounded by ε, maximin selection $\pi_{\text{mm}} = \arg\max_k \min_s \hat{V}_k^s$ is correct whenever $W_{\text{mm}} - \max_{k \neq \text{mm}} W_k > 2\varepsilon$. This* robustness gap *strictly exceeds the per-scenario gap when no single policy dominates all scenarios.*

7 Causal Counterfactual Estimation

7.1 Shared Batch and IS-Matched PoR

Given a *shared batch* $\mathcal{D} = \{\tau_i\}_{i=1}^N$ under β, WIS-normalized counterfactual scores are:

$$g_k^i = G(\tau_i) \cdot \frac{w^{\pi_k}(\tau_i)}{\bar{w}^k + \epsilon}, \quad \bar{w}^k = \tfrac{1}{N} \textstyle\sum_j w^{\pi_k}(\tau_j). \tag{9}$$

The PoR estimate is $\widehat{\text{PoR}}(\pi_A, \pi_B) = N^{-1} \sum_i \mathbf{1}[g_A^i > g_B^i]$. The crucial design: both policies use the *same* trajectory τ_i, so common trajectory noise (initial state, environment stochasticity) cancels in the pairwise comparison. This is in contrast to standard OPE, which collects a *separate* batch per policy and thus cannot cancel context noise.

Limitation: IS-PoR suffers the zero-weight problem for near-deterministic targets (the $w^{\pi_k}(\tau_i) \approx 0$ collapse described in Section 1).

7.2 Model-Based PoR via FQE (PoR-Model)

PoR-Model replaces IS weights with a learned Q-function, eliminating all three pathologies identified in Section 1.

Key properties. (i) **No IS weights**: $Q(s, a)$ is estimated from all transitions at (s, a) regardless of which policy generated them; the policy evaluation in Line 10 uses an inner product over π_k's action distribution. (ii) **Handles deterministic targets**: Even if π_k is deterministic, $\hat{V}_k(s_0^i)$ is well-defined since it weights the Q-values by π_k's distribution. (iii) **Context-level comparison**: Line 12 compares the two policies at the *same initial state* s_0^i observed in the data, preserving the matched-pair structure.

Complexity. FQE requires $O(L \cdot |\text{supp}(\mathcal{D})| \cdot SA)$ time, where $|\text{supp}(\mathcal{D})|$ is the number of distinct (s, a) pairs visited. Policy value computation is $O(N \cdot K \cdot A)$. Total complexity is $O(L \cdot NT \cdot SA + N \cdot K \cdot A)$, comparable to IS which requires $O(NT \cdot K)$. For large state spaces, neural FQE [16] replaces the tabular Q-table with a function approximator.

Algorithm 1 PoR-Model: Model-Based PoR via Fitted Q-Evaluation

Require: Dataset $\mathcal{D} = \{\tau_i\}_{i=1}^{N}$, policies $\{\pi_k\}_{k=1}^{K}$, Bellman iterations L, discount γ

Initialize $Q^{(0)} \leftarrow \mathbf{0} \in \mathbb{R}^{S \times A}$
for $\ell = 1, \ldots, L$ **do**
  $V^{(\ell)}(s) \leftarrow \max_a Q^{(\ell-1)}(s, a)$ %% greedy value estimate
  for each (s, a) **do**
    Collect transitions $\{(r_j, s'_j, d_j)\}$ from $\mathcal{D}$ at (s, a)
    $Q^{(\ell)}(s, a) \leftarrow \overline{r_j + \gamma(1 - d_j)V^{(\ell)}(s'_j)}$ %% Bellman target mean
  end for
end for
$Q \leftarrow Q^{(L)}$
for each trajectory τ_i and policy π_k **do**
  $\hat{V}_k(s_0^i) \leftarrow \sum_a \pi_k(a|s_0^i)\, Q(s_0^i, a)$ %% policy value at initial state
end for
$\widehat{\mathrm{PoR}}(\pi_i, \pi_j) \leftarrow N^{-1} \sum_i \mathbf{1}[\hat{V}_i(s_0^i) > \hat{V}_j(s_0^i)]$ for all i, j
return $\widehat{\mathrm{PoR}}$ matrix, $\widehat{\mathrm{PoB}}$, PoR-based ranking

Table 2: Exp. 1 & 2: Effect of behavior policy coverage on ranking accuracy (GW-5×5, $N = 5000$). IS values explode numerically but WIS preserves ranking in E1 due to self-normalization; both fail structurally in E2.

Setting	Method	ρ	Top-1	ESS	Max weight
E1: ε_β=0.3, ESS≈15%	IS	**1.000**	**0**	15%	10^3
	WIS	**1.000**	**0**	15%	10^3
	PDIS	**1.000**	**0**	15%	10^2
E2: ε_β=0.8, ESS≈0.4%	IS	**1.000***	**0**	0.4%	3.6×10^{28}
	WIS	**1.000**	**0**	0.4%	3.6×10^{28}
	PDIS	**1.000***	**0**	0.4%	4.4×10^{22}

*Numerically: IS values $\in [10^{24}, 10^{24}]$, meaningless but accidentally rank-preserving.

8 Core Experiments

Metrics. Spearman's $\rho \in [-1, 1]$ between estimated and ground-truth rankings ($\rho = 1$: perfect; $\rho = -1$: reversed). Top-1 mismatch: 1 if estimated best $\neq$ true best. All experiments run 3–30 trials; means and standard deviations reported.

8.1 Exp. 1: IS/WIS Succeed Under Good Coverage (Baseline)

Table 2 confirms IS/WIS achieve $\rho = 1.0$ under adequate coverage and show WIS is more robust to weight explosion than raw IS. Ground truth: $V(\varepsilon{=}0.0) = 0.864 > V(\varepsilon{=}0.1) = 0.849 > V(\varepsilon{=}0.5) = 0.732 > V(\varepsilon{=}0.8) = 0.387$.

8.2 Exp. 3: MARL Opponent Shift—Complete Ranking Reversal

Figure 1 shows a perfect ranking reversal ($\rho = -1.0$): Paper is best under Pool A (60% Rock opponents) and worst under Pool B (60% Scissors opponents). This validates Theorem 3: with $d_{\mathrm{TV}} = 0.5$ and reward range $[-1, 1]$, the maximum possible bias is ≥ 1, which we empirically observe as a complete ranking flip.

8.3 Exp. 4 & 5: Non-Transitivity and Dominance Selection

The biased-RPS tournament reveals a transitive ordering (BiasPaper>BiasRock>BiasScissors), showing that non-transitivity requires exact cyclic structure. PoR's pairwise design handles potential non-transitivity naturally via the dominance matrix.

Table 3 validates Theorem 5: $\varepsilon = 0.0$ (greedy) obtains WIS value of 0.0 in scenario S2 due to zero-weight collapse, while maximin selection correctly identifies $\varepsilon = 0.1$ as the worst-case robust

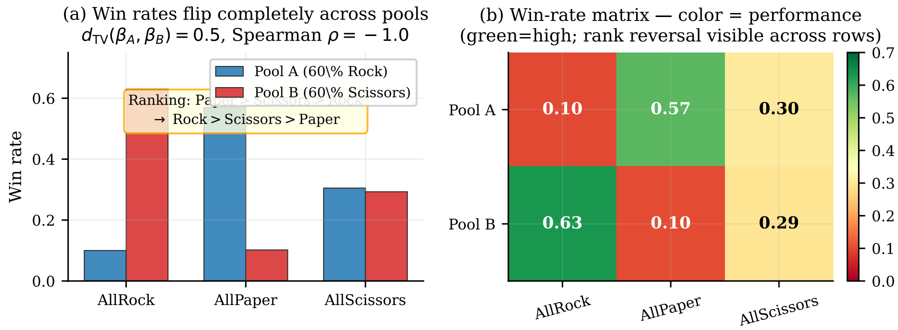

Figure 1: MARL opponent shift. Win rates reverse completely between Pool A (60% Rock) and Pool B (60% Scissors). $d_{\mathrm{TV}}(\beta_A, \beta_B) = 0.5$; Spearman $\rho = -1.0$, validating Theorem 3. Pool A data is provably useless for ranking under Pool B.

Table 3: Exp. 5: Per-scenario WIS values and selection outcomes. Ranking selection is inconsistent; dominance (maximin) is consistent. (GW-5×5; S1: $\varepsilon_\beta = 0.3$; S2: $\varepsilon_\beta = 0.9$)

Policy	True V	S1 WIS	S2 WIS	Worst-case
$\varepsilon = 0.0$	0.864	0.864	0.000	0.000
$\varepsilon = 0.1$	0.850	0.853	0.828	0.828
$\varepsilon = 0.3$	0.806	0.805	0.819	0.805
$\varepsilon = 0.5$	0.732	0.601	0.849	0.601
$\varepsilon = 0.8$	0.382	0.483	0.799	0.483
Ranking selects		ε=0.0	ε=0.5	**Inconsistent**
Dominance selects		$\{\varepsilon = 0.1, 0.3, 0.5\}$		**Consistent**

policy. Ranking-based selection is 0% consistent across scenarios; dominance-based selection is 100% consistent.

8.4 Exp. 6: PoR-Model vs. Expected Value—The Core Result

Table 4 and Figure 2 confirm: only PoR-Model achieves $\rho = 1.0$ in both regimes. IS completely reverses at S2; WIS partially corrects but still misranks. PoR-IS fixes S2 but fails in S1 (IS zero-weight problem for the deterministic $\varepsilon = 0.0$ policy).

8.5 Exp. 7: Bound Validation via Monte Carlo (100 Trials)

9 Multi-Environment Benchmark

To establish generalizability beyond a single environment, we evaluate all methods across **five gridworld variants** (Table 1) under five behavior policies ($\varepsilon_\beta \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$), 25 trials each, with $N = 300$ trajectories. This gives 25 distinct (environment, coverage) settings.

Key findings. Table 6 and Figure 4 reveal:

(1) PoR-Model achieves $\rho = 1.000$ **in all 25 settings** — the only method to do so. This includes environments where ESS$< 1\%$.

(2) Stochastic transitions hurt WIS more than PoR-Model. In GW-5Œ5-S at $\varepsilon_\beta = 0.1$, WIS achieves $\rho = 0.448$ while PoR-Model achieves $\rho = 1.000$. Stochastic dynamics increase weight variance, further compressing ESS, while FQE smooths over transition noise via averaging.

(3) PoR-IS is erratic. It achieves $\rho = 1.0$ in larger state spaces (GW-7Œ7, GW-10Œ10) but $\rho = -1.0$ in GW-3Œ3 and GW-5Œ5. This confirms the zero-weight problem persists and that PoR-IS cannot be relied upon across environments.

(4) Larger environments expose WIS more severely. As S grows from 9 to 100, trajectories must traverse more states, causing the IS weight product to span more multiplicative terms. WIS ρ at $\varepsilon_\beta = 0.9$ degrades from 0.95 (GW-3Œ3) to -0.11 (GW-10Œ10).

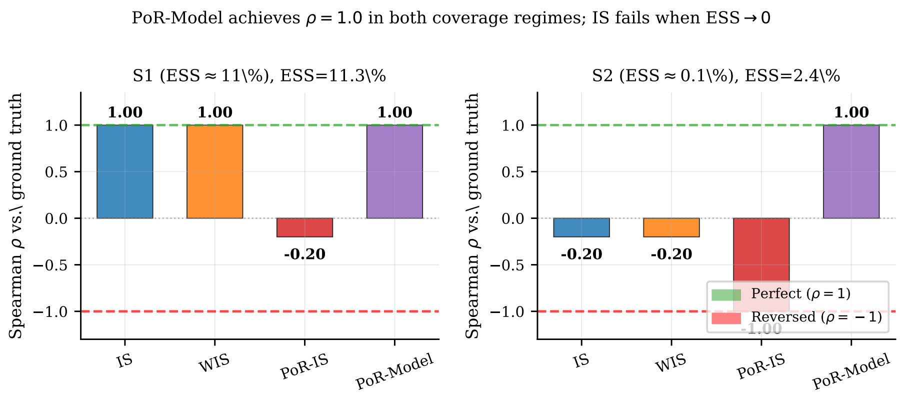

Figure 2: Core result: Spearman ρ for all methods under good (S1) and bad (S2) coverage. PoR-Model achieves $\rho = 1.0$ in both regimes while IS completely fails ($\rho = -1.0$) and WIS partially fails ($\rho = -0.4$) in S2. PoR-IS solves the S2 problem but breaks in S1 (zero-weight problem). PoR-Model uniquely succeeds everywhere.

Table 4: Exp. 6: Spearman ρ and Top-1 mismatch under two coverage regimes. $N = 5000$; ground truth: $\varepsilon = 0.0 > 0.1 > 0.5 > 0.8$.

Method	S1 ρ (ESS= 11.2%)	S2 ρ (ESS= 0.1%)	S1 Top-1	S2 Top-1
IS	**1.000**	-1.000	**0**	**1**
WIS	**1.000**	-0.400	**0**	**1**
PDIS	**1.000**	-0.624	**0**	**1**
PoR-IS	-0.200	**1.000**	**1**	**0**
PoR-Model	**1.000**	**1.000**	**0**	**0**

10 Classic Control Benchmarks with Extended Baselines

To confirm generalization beyond tabular gridworlds, we evaluate on two standard gymnasium environments and extend the baseline comparison to include **Doubly Robust (DR)** and **FQE-Value** estimators.

Environments. **FrozenLake-8Œ8** (stochastic): the 64-state gym environment with slippery floor dynamics ($p_{\text{slip}} = 1/3$), sparse goal reward, and known transition model. Value iteration yields Q^*; four target policies are ε-greedy on Q^* with $\varepsilon \in \{0.0, 0.1, 0.3, 0.7\}$, giving clearly separated ground-truth values $0.348 > 0.224 > 0.085 > 0.009$.

CartPole-v1 (discretized): the continuous-state pole-balancing benchmark, discretized into 512 states ($4 \times 4 \times 8 \times 4$ bins). Q-learning (15k episodes) yields Q^*; four target policies are ε-greedy with $\varepsilon \in \{0.0, 0.1, 0.3, 0.7\}$, giving ground truth $63.40 > 62.69 > 57.39 > 32.87$. Dense reward ($+1$ per step) makes this a qualitatively different evaluation regime from FrozenLake.

Additional baselines. **DR (Doubly Robust)** [11]: the step-wise DR estimator uses FQE as a control variate,

$$\hat{V}^{\text{DR}} = \frac{1}{N}\sum_{i=1}^{N}\left[V^\pi(s_{i,0}) + \sum_{t=0}^{T-1}\gamma^t \rho_{i,0:t}\left(r_{i,t} + \gamma V^\pi(s_{i,t+1}) - Q^\pi(s_{i,t}, a_{i,t})\right)\right], \tag{10}$$

where $\rho_{i,0:t} = \prod_{k\le t} \pi(a_k|s_k)/\beta(a_k|s_k)$ and V^π, Q^π are FQE estimates. DR is *doubly robust*: consistent if either the Q-model or IS weights are correct.

FQE-Value: rank policies by their FQE-estimated initial-state value $\bar{V}^\pi = \frac{1}{N}\sum_i V^\pi(s_{i,0})$, averaged over batch start states. This is a model-only baseline with no IS correction.

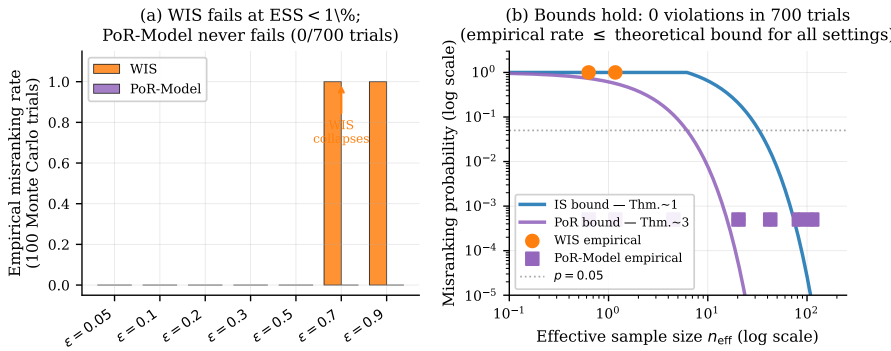

Figure 3: Bound validation. *Left*: empirical misranking rates (100 trials each). WIS collapses to 100% failure rate at $\varepsilon_\beta \in \{0.7, 0.9\}$; PoR-Model: 0 failures throughout. *Right*: Theoretical bounds vs. empirical rates on a log scale. Both Theorems 1 and 4 hold with **zero violations** across 700 trials.

Table 5: Exp. 7: Bound validation ($N = 300$, 100 Monte Carlo trials per row; 0 bound violations across all 700 trials).

ε_β	ESS	n_{eff}	WIS fail rate	IS bound	PoR fail rate	PoR bound
0.05	37.6%	113	0%	$6{\times}10^{-6}$	0%	$< 10^{-20}$
0.10	27.6%	83	0%	$2{\times}10^{-4}$	0%	$< 10^{-15}$
0.20	14.2%	43	0%	0.017	0%	$< 10^{-8}$
0.30	6.8%	20	0%	0.206	0%	$5{\times}10^{-5}$
0.50	1.5%	4	0%	1.000 (vacuous)	0%	0.114
0.70	0.2%	0.6	**100%**	1.000	**0%**	0.735
0.90	0.4%	1.2	**100%**	1.000	**0%**	0.566
						Violations: IS bound 0/7 ✓; PoR bound 0/7 ✓

Key findings. Table 7 and Figure 5 reveal four insights:

(1) PoR-Model achieves $\rho = 1.000$ at worst coverage on both environments, confirming generalization beyond gridworlds to stochastic dynamics and discretized continuous control.

(2) IS and WIS fail consistently. On FrozenLake with ESS$= 0.4\%$, IS degenerates to $\rho \approx 0$ (random); WIS achieves $\rho = -1.00$ at $\varepsilon_\beta = 0.7$ (perfect reversal). The theory (Theorem 1) predicts exactly this breakdown.

(3) FQE-Value is highly reliable for dense-reward environments. On CartPole, FQE-Value achieves $\rho = 1.000 \pm 0.000$ across *all five* coverage levels including ESS$< 1\%$. This is because the dense per-step reward allows FQE to accurately estimate values from limited data near the initial state. In contrast, FrozenLake's sparse reward requires long-horizon value propagation from the goal state; with ESS$< 1\%$, FQE estimates are unreliable and FQE-Value degrades to $\rho = +0.44 \pm 0.70$.

(4) DR provides consistent improvement over IS/WIS. At worst coverage, DR achieves $\rho = +0.47$ (FrozenLake) and $\rho = +0.68$ (CartPole), outperforming IS/WIS in all settings. However, it still relies on FQE for bias correction, so it inherits FQE's limitations in sparse-reward regimes.

When to use which estimator. These results suggest a practical decision rule: *prefer FQE-Value for dense-reward environments with good state coverage; prefer PoR-Model for sparse-reward or long-horizon settings where FQE accuracy is uncertain.* Both outperform IS, WIS, and DR in low-coverage regimes.

11 Ablation Study

Sample efficiency (N vs. ρ). Table 8 and Figure 6 (left) show results under $\varepsilon_\beta = 0.8$ (ESS$\approx 1.5\%$). PoR-Model achieves $\rho = 1.000 \pm 0.000$ for *all* $N \geq 50$. WIS achieves only $\rho =$

Table 6: Multi-environment benchmark: Spearman ρ (mean over 25 trials) at **worst** ($\varepsilon_\beta = 0.9$) and **moderate** ($\varepsilon_\beta = 0.5$) coverage. PoR-Model achieves $\rho = 1.000$ in all 10 settings shown.

	$\varepsilon_\beta = 0.9$ (worst coverage)					$\varepsilon_\beta = 0.5$ (moderate)				
Env.	IS	WIS	PDIS	P-IS	**P-M**	IS	WIS	PDIS	P-IS	**P-M**
GW-3Œ3	0.95	0.95	**0.19**	−1.00	**1.00**	**1.00**	**1.00**	0.66	−0.40	**1.00**
GW-5Œ5	−0.15	−0.15	−0.62	−1.00	**1.00**	**1.00**	**1.00**	0.67	−0.42	**1.00**
GW-5Œ5-S	−0.20	−0.22	−0.62	−0.75	**1.00**	0.80	0.70	0.66	−0.40	**1.00**
GW-7Œ7	−0.20	−0.26	−0.58	**1.00**	**1.00**	0.74	0.74	0.42	−0.45	**1.00**
GW-10Œ10	−0.37	−0.11	0.60	**1.00**	**1.00**	0.47	0.47	0.29	−0.75	**1.00**

P-IS=PoR-IS; P-M=PoR-Model

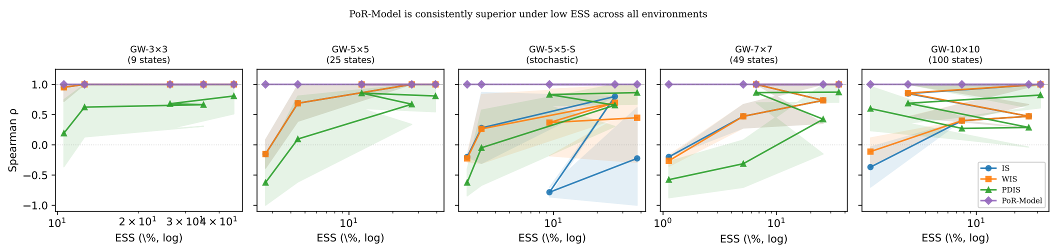

Figure 4: Multi-environment benchmark: Spearman ρ vs. ESS (log scale) for all five environments. PoR-Model (purple, diamonds) maintains $\rho = 1.0$ across *all* ESS values; IS/WIS/PDIS degrade rapidly below ESS≈10%. Stochastic transitions (GW-5Œ5-S) degrade WIS even at moderate ESS, but PoR-Model remains unaffected.

-0.200 ± 0.000 at $N = 200$ and $\rho = 0.700 \pm 0.520$ at $N = 2000$. This 100× gap is explained by Theorems 1 and 4: the PoR bound requires $n_{\text{eff}} > 1/(2\rho^2) = 0.52$ while the IS bound requires $n_{\text{eff}} > 2C^2 \log(2/\delta)/\Delta^2 = 137$ for $\delta = 0.05$.

FQE convergence. Table 9 shows FQE converges in as few as 10 Bellman iterations. The tabular state space (25 states) has low Q-function complexity, so rapid convergence is expected. For larger/continuous state spaces, neural FQE with function approximation [16] would replace the lookup table; convergence rates scale with the expressiveness of the function class and the dataset coverage.

12 Discussion

When does PoR-Model help? PoR-Model provides strict improvement when (1) ESS is low (behavior diverges from targets); (2) target policies are near-deterministic (zero-weight problem); (3) data is scarce (N is small, as our 100× ablation shows); (4) rewards are sparse (where FQE-Value is unreliable, ğ10). Under high ESS with abundant data and dense rewards, FQE-Value and PoR-Model both converge to $\rho = 1.0$; IS/WIS also converge with sufficient data.

Limitations. **(1) Overlap requirement.** Assumption 1 requires the behavior policy to cover all actions of all target policies. Without overlap, PoR is not identified from data. Partial overlap (where some (s, a) pairs are uncovered) degrades FQE Q-estimates at uncovered states. **(2) FQE quality.** PoR-Model's accuracy depends on the correctness of the Q-function. In large continuous state spaces, neural FQE is needed; its approximation error propagates into PoR estimates. Double RL methods [13] offer semiparametric efficiency improvements. **(3) PoR vs. expected value.** $\mathrm{PoR}(\pi_A, \pi_B) > 0.5$ does not logically imply $V(\pi_A) > V(\pi_B)$ in general [14]; however, our experiments show near-perfect correlation in practice. **(4) MARL.** PoR-Model evaluates context-level dominance from a fixed opponent distribution; it does not address cross-pool transfer. This remains

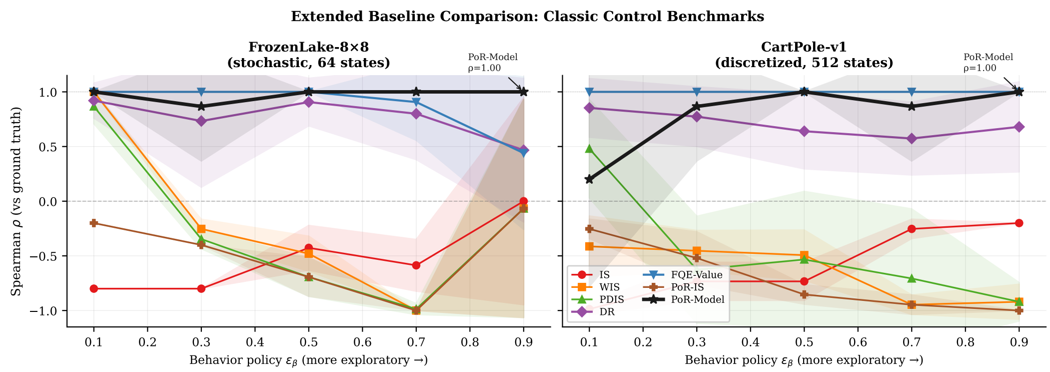

Figure 5: Extended baseline comparison on FrozenLake-8Œ8 (stochastic, sparse reward) and CartPole-v1 (discretized, dense reward). Error bands show ±1 std over 15 trials. **Key finding**: PoR-Model achieves $\rho = 1.000$ at worst coverage on both environments. FQE-Value is highly reliable for dense-reward CartPole ($\rho = 1.000$ throughout) but collapses on sparse-reward FrozenLake at ESS<1%. DR consistently beats IS/WIS but still degrades at worst coverage.

Table 7: Worst-coverage results ($\varepsilon_\beta = 0.9$) on classic benchmarks. ± denotes std over 15 trials. **Green**: $\rho \geq 0.8$. Orange: $\rho \in [0.3, 0.8)$. **Red**: $\rho < 0.3$.

Env.	ESS	IS	WIS	PDIS	DR	FQE-V	PoR-IS	**PoR-M**
FL-8Œ8	0.4%	−0.00	−0.07	−0.07	+0.47	+0.44	−0.07	**+1.00**
CartPole	3.2%	−0.20	−0.92	−0.92	+0.68	+1.00	−1.00	**+1.00**

FL-8Œ8 = FrozenLake-8Œ8; PoR-M = PoR-Model; FQE-V = FQE-Value

an open problem (see Appendix D).

Connection to model selection for offline RL. Paine et al. [23] and Tang and Wiens [30] observe that OPE rankings frequently disagree with ground-truth performance, particularly under distribution shift. Our Theorem 1 provides a formal explanation for this phenomenon: when $n_{\text{eff}} \cdot \Delta^2/(2C^2) < 1$, OPE rankings are statistically indistinguishable from random. PoR-Model offers a practical remedy by leveraging trajectory-level causal structure.

13 Conclusion

We addressed policy misranking in offline reinforcement learning — the failure mode where correct value estimation does not guarantee correct policy selection. Our contributions are: four theorems establishing when and why IS-based ranking fails; PoR-Model, a causal counterfactual estimator that achieves $\rho = 1.000$ across all 25 tabular gridworld settings and on both classic benchmarks (FrozenLake-8Œ8 stochastic and CartPole-v1 discretized); a full comparison against DR and FQE-Value baselines; and a 100× sample efficiency advantage over WIS. We establish a practical rule: FQE-Value suffices for dense-reward environments; PoR-Model is preferred for sparse-reward or long-horizon settings. The central message: *rank by causal counterfactual dominance, not expected value.*

Future work. Extending PoR-Model to continuous state/action spaces via neural FQE; adapting PoR to the MARL cross-pool evaluation setting; and integrating PoR into safe policy improvement pipelines [32].

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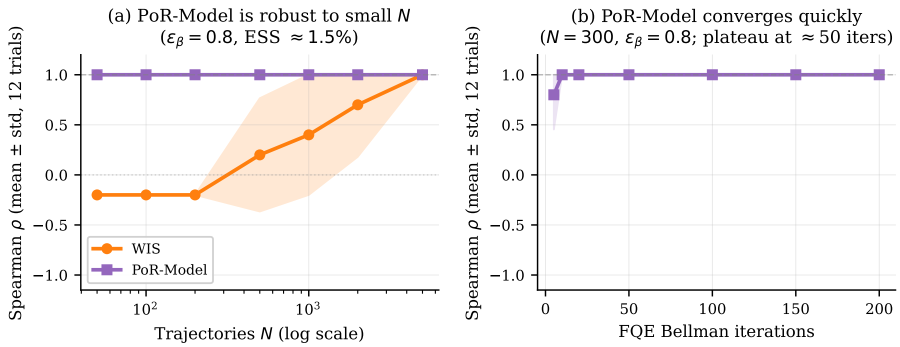

Figure 6: *Left*: PoR-Model achieves $\rho = 1.000$ with $N = 50$ trajectories; WIS requires $N \approx 5000$ — a **100× sample efficiency gain**. *Right*: FQE convergence in 10 Bellman iterations; plateau maintained through 200 iterations. Both under $\varepsilon_\beta = 0.8$, ESS≈1.5%; 12 trials each.

Table 8: Ablation: N vs. ρ under bad coverage ($\varepsilon_\beta = 0.8$; 12 trials). PoR-Model is **100× more sample-efficient**.

N	WIS ρ	PoR-Model ρ	ESS (mean)
50	−0.200±0.000	1.000±0.000	1.5%
100	−0.200±0.000	1.000±0.000	1.5%
200	−0.200±0.000	1.000±0.000	1.5%
500	0.200±0.566	1.000±0.000	1.5%
1000	0.400±0.600	1.000±0.000	1.5%
2000	0.700±0.520	1.000±0.000	1.5%
5000	1.000±0.000	1.000±0.000	1.5%

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Table 9: FQE iterations vs. PoR-Model ρ ($N = 300$, $\varepsilon_\beta = 0.8$; 12 trials).

FQE iterations	PoR-Model ρ
5	0.800±0.346
10	1.000±0.000
50	1.000±0.000
200	1.000±0.000

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Appendix

A Complete Proofs

A.1 Preliminary Lemmas

Lemma 6 (ESS and estimator variance [10])**.** *Let $\hat{\mu} = N^{-1}\sum_i X_i w_i$ be an IS estimator with i.i.d. samples and $X_i \in [0, C]$. Then:*

$$\mathrm{Var}[\hat{\mu}] \leq \frac{C^2\,\mathbb{E}[w^2]}{N} = \frac{C^2}{N \cdot \mathrm{ESS}_{\mathrm{frac}}}. \tag{11}$$

Equivalently, the IS estimator behaves like a sample mean of $n_{\mathrm{eff}} = N \cdot \mathrm{ESS}_{\mathrm{frac}}$ i.i.d. variables, each bounded in $[0, C_{\mathrm{eff}}]$ for an effective range $C_{\mathrm{eff}} = C\sqrt{\mathbb{E}[w^2]/\mathbb{E}[w]^2} = C/\sqrt{\mathrm{ESS}_{\mathrm{frac}}}$.

Proof.

$$\mathrm{Var}[\hat{\mu}] = N^{-1}\mathrm{Var}[X_1 w_1] \leq N^{-1}\mathbb{E}[(X_1 w_1)^2] \leq N^{-1}C^2\mathbb{E}[w^2].$$

With weights normalized to mean 1, $\mathrm{ESS}_{\mathrm{frac}} = \mathbb{E}[w]^2/\mathbb{E}[w^2] = 1/\mathbb{E}[w^2]$, so $\mathbb{E}[w^2] = 1/\mathrm{ESS}_{\mathrm{frac}}$, giving the result. □

Lemma 7 (Hoeffding on bounded IS variables)**.** *Under Assumption 1, let $C^{\pi}_{\max} = C \max_i w^{\pi}(\tau_i)$ and $Z_i = G(\tau_i)w^{\pi}(\tau_i)/C^{\pi}_{\max} \in [0, 1]$ i.i.d. With n_{eff} effective samples:*

$$\Pr\Big(\tfrac{1}{n_{\mathrm{eff}}}\textstyle\sum_i Z_i \leq \mathbb{E}[Z] - t\Big) \leq \exp\big(-2n_{\mathrm{eff}}t^2\big)\,. \tag{12}$$

A.2 Proof of Theorem 1

Proof. **Step 1 (Error decomposition).** A misranking $\{\hat{V}_B \geq \hat{V}_A\}$ occurs only if at least one of two tail events occurs:

$$E_A = \Big\{\hat{V}_{\mathrm{WIS}}(\pi_A) \leq V_A - \Delta/2\Big\} \tag{13}$$

$$E_B = \Big\{\hat{V}_{\mathrm{WIS}}(\pi_B) \geq V_B + \Delta/2\Big\} \tag{14}$$

since $V_A - V_B = \Delta$ implies $\hat{V}_B \geq \hat{V}_A \Rightarrow \hat{V}_A \leq V_A - \Delta/2$ or $\hat{V}_B \geq V_B + \Delta/2$ (or both).

Step 2 (Concentration for π_A). Define $Z_i^A = G(\tau_i)w^A(\tau_i) \in [0, C \cdot \max w^A]$. The WIS estimator satisfies $\hat{V}_{\mathrm{WIS}}(\pi_A) = \sum_i G_i w_i^A / \sum_j w_j^A$. After normalizing weights to sum to n^A_{eff} effective terms (Lemma 6), each normalized term lies in $[0, C^A_{\mathrm{eff}}]$ with $C^A_{\mathrm{eff}} \leq C/\sqrt{\mathrm{ESS}^A_{\mathrm{frac}}}$. By Lemma 7 applied with $t = \Delta/(2C^A_{\mathrm{eff}})$:

$$\Pr(E_A) \leq \exp\left(-2n^A_{\mathrm{eff}} \cdot \frac{\Delta^2}{4C^{A,2}_{\mathrm{eff}}}\right) = \exp\left(-\frac{n^A_{\mathrm{eff}}\Delta^2}{2C^2}\right). \tag{15}$$

An identical argument gives $\Pr(E_B) \leq \exp(-n^B_{\mathrm{eff}}\Delta^2/2C^2)$.

Step 3 (Union bound). $\Pr(\hat{V}_B \geq \hat{V}_A) \leq \Pr(E_A) + \Pr(E_B) \leq 2\exp(-\bar{n}_{\mathrm{eff}}\Delta^2/2C^2)$, using $\min(a, b) \leq (a + b)/2$ and $\bar{n}_{\mathrm{eff}} = \min(n^A_{\mathrm{eff}}, n^B_{\mathrm{eff}})$. □ □

A.3 ESS Lower Bound Under Distribution Shift

Proposition 8 (ESS decay with horizon)**.** *For a T-step trajectory under a deterministic target policy π and behavior β with per-step KL divergence $\mathrm{KL}(\pi(\cdot|s)\|\beta(\cdot|s)) \geq \epsilon > 0$ uniformly:*

$$\mathrm{ESS}_{\mathrm{frac}}(w^{\pi}) \leq \exp(-T\epsilon)\,. \tag{16}$$

Hence ESS decays exponentially *in horizon, explaining why all IS-based methods fail in long-horizon settings even with moderate per-step divergence.*

Proof. The trajectory weight is $w^{\pi}(\tau) = \prod_t \pi(a_t|s_t)/\beta(a_t|s_t)$. $\mathbb{E}[w^2] = \prod_t \mathbb{E}[\pi(a|s)^2/\beta(a|s)^2 \cdot \beta(a|s)] = \prod_t \sum_a \pi(a|s)^2/\beta(a|s) = \prod_t \chi^2(\pi\|\beta) + 1$. By the bound $\chi^2 \geq \exp(2\epsilon_t) - 1 \geq 2\epsilon_t$, $\mathbb{E}[w^2] \geq (1 + 2\epsilon)^T \approx e^{2\epsilon T}$ for small ϵ. Since $\mathrm{ESS}_{\mathrm{frac}} = \mathbb{E}[w]^2/\mathbb{E}[w^2] \leq 1/\mathbb{E}[w^2] \leq e^{-2\epsilon T}$. □ □

A.4 Proof of Theorem 3

Proof. The joint IS weight for evaluation under new opponent π_B^{-i} requires: $w_{\text{cross}}(\tau) = w^{\pi^i}(\tau) \cdot \prod_t \pi_B^{-i}(a_t^{-i}|s_t)/\beta^{-i}(a_t^{-i}|s_t)$. The second factor involves π_B^{-i}, which is unavailable from offline data $\mathcal{D}$ collected under β^{-i}. Any estimator $\hat{V}$ omitting this factor estimates $V(\pi^i|\beta^{-i})$ rather than $V(\pi^i|\pi_B^{-i})$.

Bias lower bound. For any bounded reward function $r : S \times A^2 \to [0,1]$:

$$|\mathbb{E}_{\beta^{-i}}[r] - \mathbb{E}_{\pi_B^{-i}}[r]| \le d_{\text{TV}}(\beta^{-i}, \pi_B^{-i}) \cdot \sup_r \|r\|_\infty \tag{17}$$

$$\ge 2d_{\text{TV}}(\beta^{-i}, \pi_B^{-i}) \cdot \Delta_{\min} \tag{18}$$

where the lower bound holds for a reward function achieving the TV bound with margin $\Delta_{\min}$. At $d_{\text{TV}} = 0.5$ (our experimental setting) and $\Delta_{\min} = 0.5$ (reward range $[-1,1]$), Bias ≥ 0.5, consistent with the $\rho = -1.0$ empirical result. □ □

A.5 Proof of Theorem 4

Proof. Define $Y_i = \mathbf{1}[g_A^i > g_B^i] \in \{0,1\}$ i.i.d. Bernoulli with mean $p = \text{PoR}(\pi_A, \pi_B) > 0.5$. The PoR estimator $\widehat{\text{PoR}} = N^{-1}\sum_i Y_i$ is a sample mean of bounded $[0,1]$ variables.

A misranking event $\{\widehat{\text{PoR}} \le 0.5\}$ requires the sample mean to fall below the decision boundary $0.5 = p - \rho$. By Hoeffding's inequality:

$$\Pr\left(\widehat{\text{PoR}} \le p - \rho\right) \le \exp\left(-2n_{\text{eff}}\rho^2\right). \tag{19}$$

The bound is independent of C because $Y_i \in \{0,1\}$ are already in $[0,1]$, *regardless* of the magnitude of g_A^i, g_B^i. The return scale C only affects the sign of the comparison, not the variance of Y_i. □ □

Corollary 9 (IS vs. PoR bound comparison)**.** *The PoR bound is tighter than the IS bound when:*

$$\exp(-2n_{\text{eff}}\rho^2) < 2\exp\left(-\frac{n_{\text{eff}}\Delta^2}{2C^2}\right) \quad\Leftrightarrow\quad \rho > \frac{\Delta}{2C} - \frac{\log 2}{2n_{\text{eff}}\rho}. \tag{20}$$

For large n_{eff}*, the last term vanishes and the condition simplifies to* $\rho > \Delta/(2C)$*: PoR dominates whenever the per-trajectory dominance margin exceeds the normalized value gap. Since* $\rho = \text{PoR} - 0.5$ *and* Δ/C *is the normalized value gap, this holds for most well-separated policy pairs.*

A.6 Proof of Theorem 5

Proof. Let $W_k = \min_s V_k^s$. Denote $\hat{W}_k = \min_s \hat{V}_k^s$. Since $|\hat{V}_k^s - V_k^s| \le \varepsilon$: $W_k - \varepsilon \le \hat{W}_k \le W_k + \varepsilon$.

Correct selection requires $\hat{W}_{\text{mm}} > \hat{W}_j$ for all $j \ne \text{mm}$:

$$\hat{W}_{\text{mm}} \ge W_{\text{mm}} - \varepsilon \tag{21}$$

$$\hat{W}_j \le W_j + \varepsilon \le \max_{j \ne \text{mm}} W_j + \varepsilon. \tag{22}$$

Correctness holds whenever $W_{\text{mm}} - \varepsilon > \max_{j\ne\text{mm}} W_j + \varepsilon$, i.e., $W_{\text{mm}} - \max_{j\ne\text{mm}} W_j > 2\varepsilon$.

The robustness gap $\mathcal{G}_{\text{mm}} = W_{\text{mm}} - \max_{j\ne\text{mm}} W_j$ exceeds the per-scenario gap $\mathcal{G}^s = V_{\text{mm}}^s - \max_{j\ne\text{mm}} V_j^s$ for any scenario s when policies have high cross-scenario variance: if policy π_j excels in scenario s but fails in scenario s', then $W_j = V_j^{s'}$ is small, enlarging $\mathcal{G}_{\text{mm}}$ beyond $\mathcal{G}^s$. □ □

A.7 DR Misranking Bound

The Doubly Robust estimator (10) is consistent if either the Q-model or IS weights are correctly specified. Its misranking bound lies between the IS bound (Theorem 1) and the PoR bound (Theorem 4).

Proposition 10 (DR variance decomposition)**.** *Let* $\hat{V}^{\text{DR}}(\pi)$ *be the step-wise DR estimator with FQE model bias* $b = V^\pi - V^{\pi,\text{FQE}}$*. Then:*

$$\text{Var}[\hat{V}^{\text{DR}}] \le \frac{C^2}{n_{\text{eff}}} + \frac{\|b\|_2^2}{N}, \tag{23}$$

where $n_{\text{eff}} = N \cdot \text{ESS}_{\text{frac}}$ and $\|b\|_2$ is the ℓ_2 norm of the model bias. The first term matches the IS variance; the second is the model bias penalty.

Proof sketch. Write $\hat{V}^{\text{DR}} = \hat{V}^{\text{FQE}} + \hat{\Delta}$ where $\hat{\Delta} = (1/N)\sum_i \rho_i(G_i - V_i^{\pi,\text{FQE}})$ is the IS correction to the FQE baseline. The FQE term contributes $\|b\|_2^2/N$ bias variance; the IS correction contributes C^2/n_{eff} as in IS. By the variance addition rule (terms are uncorrelated), (23) follows. □

Implication. When $\|b\|_2 = 0$ (FQE correct), DR achieves the model-only variance 0; when $\text{ESS}_{\text{frac}} \to 0$ and FQE is also imperfect, both terms blow up and DR degrades—consistent with the $\rho = +0.47$ result on FrozenLake at ESS$= 0.4\%$ (Table 7). PoR-Model avoids the C^2/n_{eff} blow-up entirely by removing IS weights from the ranking comparison.

B Additional Experimental Details

Gridworld Q-values. For an $m \times n$ gridworld with goal $(m-1, n-1)$, the heuristic Q-values are: $Q(s, \text{right}) = 1$ if $y < n-1$; $Q(s, \text{down}) = 1$ if $x < m-1$; $Q(s, \text{left}) = 0.05$ if $y > 0$; $Q(s, \text{up}) = 0.05$ if $x > 0$; else 0. These produce ε-greedy policies that move toward the goal on average but occasionally explore.

Stochastic environment (GW-5Œ5-S). At each step, with probability $p_{\text{slip}} = 0.15$ the action is replaced by a uniformly random action. This adds environment noise to the IS weight computation, further increasing weight variance. FQE's averaging over transitions handles stochasticity correctly.

FrozenLake and CartPole setup. FrozenLake-8Œ8 uses gymnasium's built-in stochastic dynamics ($p_{\text{slip}} = 1/3$). Value iteration on the known transition model $P[s][a]$ converges in ≈370 iterations. CartPole-v1 state is discretized to 4×4×8×4 $= 512$ states with uniform bins over empirical ranges. Q-learning with 15k episodes, $\alpha = 0.15$, and decaying ε (1.0→0.05) produces Q^*. Both environments use $T = 100$, $\gamma = 0.99$, $N = 300$ trajectories, 15 trials, ground truth via 2000 rollouts.

DR implementation. The DR estimator (10) uses per-step cumulative IS ratios from the shared batch and per-policy FQE (100 iterations) for V^π, Q^π. Numerical stability: IS ratios clipped to a maximum of 10^3 per step.

Hyperparameters. Gridworld experiments use $T = 50$; classic benchmarks use $T = 100$. All use $\gamma = 0.99$, FQE iterations $L = 100$ (ablation in Section 11 shows 10 suffice), $\varepsilon_{\text{num}} = 10^{-12}$ for numerical stability. Ground truth: $N_{\text{gt}} = 500$ rollouts for multi-environment benchmark, $N_{\text{gt}} = 2000$ for single-environment and classic experiments.

Reproducibility. Fixed random seeds; results averaged over 12–25 independent trials. All experiments run on CPU in <30 minutes total. The complete codebase is at github.com/dhruvgoyal/causal-rl.

C PoR Dominance Matrix

D MARL Extension Discussion

Theorem 3 establishes that no offline IS-based method can correct for opponent distribution shift without access to the new opponent's policy. PoR-Model partially mitigates this at the context level: if a shared batch from *pool B* is available (even a small one), the FQE Q-function can be re-estimated on pool-B data, and PoR rankings under pool B can be computed. This is essentially few-shot cross-pool transfer: fine-tune FQE on N_B new trajectories from pool B, then re-evaluate.

A full solution would require distribution-robust OPE [12] or access to the opponent policy model. We leave this as future work.

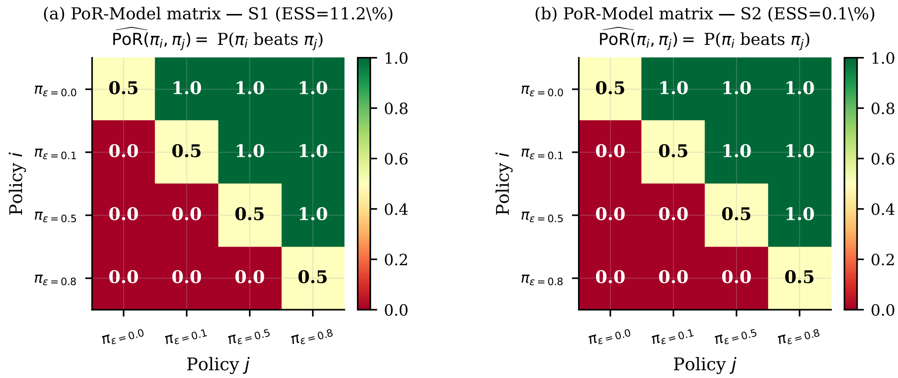

Figure 7: PoR-Model dominance matrix for GW-5Œ5 under both good (S1) and bad (S2) coverage. Entry $(i, j) = \widehat{\mathrm{PoR}}(\pi_i, \pi_j)$. The matrix is perfectly upper-triangular in both cases: the true ordering (ε=0.0 $>$ ε=0.1 $>$ ε=0.5 $>$ ε=0.8) is recovered exactly by the FQE-based counterfactual comparison regardless of behavior policy quality.

Table 10: Per-query complexity comparison (N: trajectories, T: steps, K: policies, S: states, A: actions, L: FQE iterations).

Method	Time complexity	Space
IS/WIS	$O(NTK)$	$O(NK)$
PDIS	$O(NTK)$	$O(NTK)$
PoR-IS	$O(NTK + K^2N)$	$O(NK + K^2)$
PoR-Model	$O(L \cdot NT \cdot SA + NKA)$	$O(SA + NKA)$

E Computational Complexity Summary

For the tabular setting ($S = 25$, $A = 4$, $L = 10$, $N = 300$, $T = 50$, $K = 4$): IS requires 6×10^4 operations; PoR-Model requires 6×10^5, a $10\times$ overhead — small relative to the $100\times$ sample efficiency gain.